Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page.

Name: _____

Student ID: _____

Quiz 3: 09/15 Content

MATH 1271

09/17/2026

This quiz is worth 10 points and consists of 1 questions. You have 5 minutes to complete the quiz.

Question	Points
1	10
Total:	10

1. Evaluate the given limits of the piecewise defined function.

$$f(x) = \begin{cases} x+3, & x \leq 0 \\ x^2+2, & x > 0 \end{cases} \tag{1}$$

(a) (3 points) $\lim_{x\to 0^-} f(x)$

(b) (3 points) $\lim_{x\to 0^+} f(x)$

(c) (3 points) $\lim_{x\to 0} f(x)$

(d) (1 point) $f(0)$

Solution:

1. $\lim\limits_{x\to 0^-} f(x) = 3$ since when $x < 0$, $f(x) = x + 3$ which is continuous. Thus $\lim\limits_{x\to 0^-} f(x) = \lim\limits_{x\to 0^-} x + 3 = 0 + 3 = 0$
2. $\lim\limits_{x\to 0^+} f(x) = 2$ by the same reasoning since when $x > 0$, $f(x) = x^2 + 2$.
3. $\lim\limits_{x\to 0} f(x)$ DNE since the one sided limits are not equal to eachother.
4. By the piecewise definition, when $x = 0$, $f(x) = x + 3$. So $f(0) = 0 + 3 = 3$.